

**B.E. II EXAMINATION DEC.' 2023**  
**Information Technology**  
**AIR3C1: Discrete Mathematics**

Duration: 3 Hrs]

[Max. Marks: 60

Note: Attempt any **TWO** parts from each question. Each carries equal marks.

Q1. (a) Determine the number of integers between **1 and 350** that are divisible by any of the integers **2, 5, 8 and 11**. (06)

(b) For the following hashing function, show how the following data would be inserted in the order given in initially empty cells. Use the usual collision resolution policy to resolve collision, if occurs:  $h(x) = (x^2 + x) \pmod{13}$ , cells indexed from 0 to 12; the data is: **142, 29, 821, 378, 1048, 2965, 1032, 1983, 34, 23**. (06)

(c) Construct the Hasse diagram for the partial ordering  $\{(a, b): a \text{ divides } b\}$  on the set  $\{2, 3, 5, 9, 12, 15, 18, 24, 36, 45\}$ , writing all steps properly. (06)

Q2. (a) Check whether the following is a tautology or not:  $[(p \vee q) \wedge (\sim p \vee r)] \rightarrow (q \vee r)$  (06)

(b) State the converse, contrapositive and inverse of the following implication: (06)

*"To get the tenure as a Professor, it is sufficient to be world famous".*

(c) (i) Translate the following into a logical expression using predicates, quantifiers and logical connectives: (03)

*"One of your tool is not in the correct place, but it is in excellent condition".*

Then form the negation and again express as a sentence.

(ii) Let  $S(x)$  be the predicate " $x$  is a student,"  $F(x)$  the predicate " $x$  is a faculty member," and  $A(x, y)$  the predicate " $x$  has asked  $y$  a question," where the domain consists of all people associated with your school. Use quantifiers to express each of these statements: (03)

1) *Lois has asked Professor Michaels a question.*

2) *Some student has not asked any faculty member a question.*

3) *There is a faculty member who has never been asked a question by a student.*

Q3. (a) Using mathematical induction prove that  $1^3 + 2^3 + 3^3 + \dots + n^3 = \left[ \frac{n(n+1)}{2} \right]^2$ ,  $\forall n$ . (06)

(b) Use generating function to solve the recurrence relation  $a_n = 8a_{n-1} + 10^{n-1}$ , where  $a_0 = 1$ . (06)

(c) Find the solution to the recurrence relation  $a_n + 5a_{n-1} + 6a_{n-2} = 3r^2 - 2r + 1$ . (06)

Q4. (a) Find the length of a shortest path between  $a$  and  $z$  in the given weighted graph, using Dijkstra's algorithm: (06)

